

2007 普物

一、

$$1. \vec{v} = 10t\vec{i} + 10t^2\vec{j} \quad v = \sqrt{100 + 100t^2} = 10\sqrt{1+t^2} \quad a = \frac{dv}{dt} = 10 \cdot 2t \cdot (1+t^2)^{-\frac{1}{2}} = \frac{20t}{\sqrt{1+t^2}}$$

$$a_n = 3\sqrt{2} \text{ m/s}$$

$$2. \Delta p = F \cdot t = 200 \Rightarrow W = \Delta E_n = \frac{(\Delta p)^2}{2m} = \frac{200^2}{2 \times 5} = 4000$$

$$3. \Delta \varphi = \pi$$

$$4. \nu_1 = \frac{330}{330-2} \times 400 = 392.6 \text{ Hz} \quad \nu_2 = \frac{330}{330+2} \times 400 = 397.4 \text{ Hz} \Rightarrow |\nu| = 4.8 \text{ Hz}$$

5.

$$6. \nu_{C_{12}H_6} = 2 \quad \nu_{C_{12}H_6} = \frac{3}{2}R \quad \nu_{C_{12}H_6} = \frac{5}{2}R \quad \therefore \Delta H_e = 3J$$

$$7. PV = \nu RT = \frac{m}{M} RT \Rightarrow P = \rho \cdot \frac{RT}{M}$$

$$\therefore \sqrt{P} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3 \times 10^2 \times 1.01 \times 10^3}{124 \times 10^{-2}}} = 4.943 \times 10^3 \text{ m/s}$$

$$8. E_0 \cdot d = A, E = \frac{\sigma}{\epsilon_0} \Rightarrow \sigma = \frac{2\epsilon_0 A}{d}$$

$$9. 2d(n-1) = \lambda n \Rightarrow 2 \times 100 \times 10^{-3} \times (n-1) = 100 \times 589 \text{ nm} \Rightarrow n = 1.003$$

$$10. d = \frac{x}{4(n_e - n_o)} = 1.23 \times 10^{-5} \text{ m}$$

$$11. T\lambda_m = b \Rightarrow \frac{T}{\lambda} = \frac{\lambda_m}{\lambda_0}, M = \sigma T^4 \Rightarrow \frac{\lambda_0}{\lambda} = \frac{T_0}{T} = \left(\frac{\lambda_m}{\lambda_0}\right)^4 = 7.63 \text{ 倍}$$

$$12. E = -\frac{13.6 \text{ eV}}{2} = -3.4 \text{ eV}, L = \sqrt{l(l+1)}\hbar$$

二、

1. ① 欧姆：欧姆定律。 ② 法拉第：电磁感应定律
③ 麦克斯韦：麦克斯韦方程组 ④ 楞次：楞次定律

$$2. \begin{cases} F - f = Ma_c \\ Rf - rF = J\beta \\ a_c = R\beta \end{cases} \Rightarrow \begin{cases} 25 - f = fa_c \\ 0.2f - 0.1f = 2.0\beta \\ a_c = 2\beta \end{cases} \Rightarrow \begin{cases} f = 17N \\ \beta = 10 \text{ rad/s}^2 \\ a_c = 20 \text{ m/s}^2 \end{cases}$$

$$17 \leq \mu mg \Rightarrow \mu \geq \frac{1}{60} = 0.025$$

$$3. (1) D = 2\pi r = \lambda, D = \epsilon_0 \epsilon_r E \Rightarrow E = \frac{2}{2\pi \epsilon_0 \epsilon_r} \cdot \frac{1}{r}$$

$$\therefore U = \int_{R_1}^{R_2} E dr = \frac{2}{2\pi \epsilon_0 \epsilon_r} \ln \frac{R_2}{R_1} \Rightarrow \lambda = \frac{2\pi \epsilon_0 \epsilon_r}{\ln \frac{R_2}{R_1}}$$

$$(2) E = \frac{30}{4\pi \times 10^{-2}} \times \frac{1}{0.035} = 1000 \text{ N/C}$$

$$(3) 0.4 = \frac{U}{2\pi \epsilon_0} \cdot \ln \frac{R_2}{R_1} = \frac{30}{4\pi \times 10^{-2}} \cdot \ln \frac{R_2}{0.035} = 12.46 \text{ V}$$

$$4. (1) F_m = R X_m \Rightarrow 0.8 = R \cdot 0.4 \Rightarrow R = 2 \text{ N/m}; \quad \frac{1}{2} m \omega^2 = \frac{1}{2} R X_m^2 \Rightarrow m = R \cdot \left(\frac{X_m}{\omega}\right)^2 = 2 \times \left(\frac{0.4}{0.8\pi}\right)^2 = \frac{1}{\pi^2}$$

$$\therefore E = \frac{1}{2} \times m \times \omega^2 = \frac{1}{2} \times \frac{1}{\pi^2} \times (0.8\pi)^2 = 0.16 \text{ J}$$

$$(2) T = 2\pi \sqrt{\frac{m}{R}} = 2\pi \sqrt{\frac{1}{2\pi^2}} = 1 \quad \therefore \omega = \frac{2\pi}{T} = 2\pi$$

$$\therefore x = 0.4 \cos\left(2\pi t + \frac{\pi}{2}\right)$$

$$5. (1) f(v) = \begin{cases} v^2, & 0 \leq v \leq v_0 \\ 0, & v > v_0 \end{cases} \Rightarrow \begin{array}{c} f(v) \\ v_0^2 \\ \nearrow \\ 0 \quad v_0 \end{array}$$

$$(2) \int_0^{v_0} A v^2 dv = 1 \Rightarrow A \cdot \frac{v^3}{3} = 1 \Rightarrow A = \frac{3}{v_0^3}$$

$$(3) v_p = v_0, \quad \bar{v} = \int_0^{v_0} v \cdot f(v) dv = \int_0^{v_0} \frac{3}{v_0^3} v^3 dv = \frac{3}{v_0^3} \cdot \frac{v_0^4}{4} = \frac{3}{4} v_0$$

$$\sqrt{\bar{v}} = \sqrt{\int_0^{v_0} \frac{3}{v_0^3} v^3 dv} = \sqrt{\frac{3}{5} v_0^2}$$

$$(4) \overline{v^2} = \frac{\int_0^{v_0} v^2 \cdot f(v) dv}{\int_0^{v_0} f(v) dv} = \frac{\int_0^{v_0} \frac{3}{v_0^3} v^4 dv}{\int_0^{v_0} \frac{3}{v_0^3} v^2 dv} = \frac{3 v_0^2}{20} \quad \sqrt{\bar{v^2}} = \sqrt{\frac{3 v_0^2}{20}}$$

$$6. (1) -\frac{\pi}{2} \leq \sin \theta \leq \frac{\pi}{2}, \quad d \sin \theta = k \lambda \Rightarrow k = \frac{d}{\lambda} \sin \theta = \frac{d}{\lambda} \cdot \frac{\lambda}{2} = 5 \quad \therefore 9 \text{ 条}: 0, \pm 1, \pm 2, \pm 3, \pm 4$$

$$(2) \Delta x = \frac{\lambda}{2} = \frac{2 \times 440 \text{ nm}}{2 \times 4 \times 10^3} = 2.4 \times 10^{-3} \text{ m}$$

$$7. \phi = 135 = \int_0^{0.2} 4t^2 y \cdot 0.2 dy = 0.8 t^2 \left(\frac{y^2}{2}\right) \Big|_0^{0.2} = 0.016 t^2$$

$$\therefore E = -\frac{\partial \phi}{\partial t} = -0.032 t = -0.032 \times 0.025 = 0.008 \text{ V/C} \quad \text{方向为顺时针方向.}$$